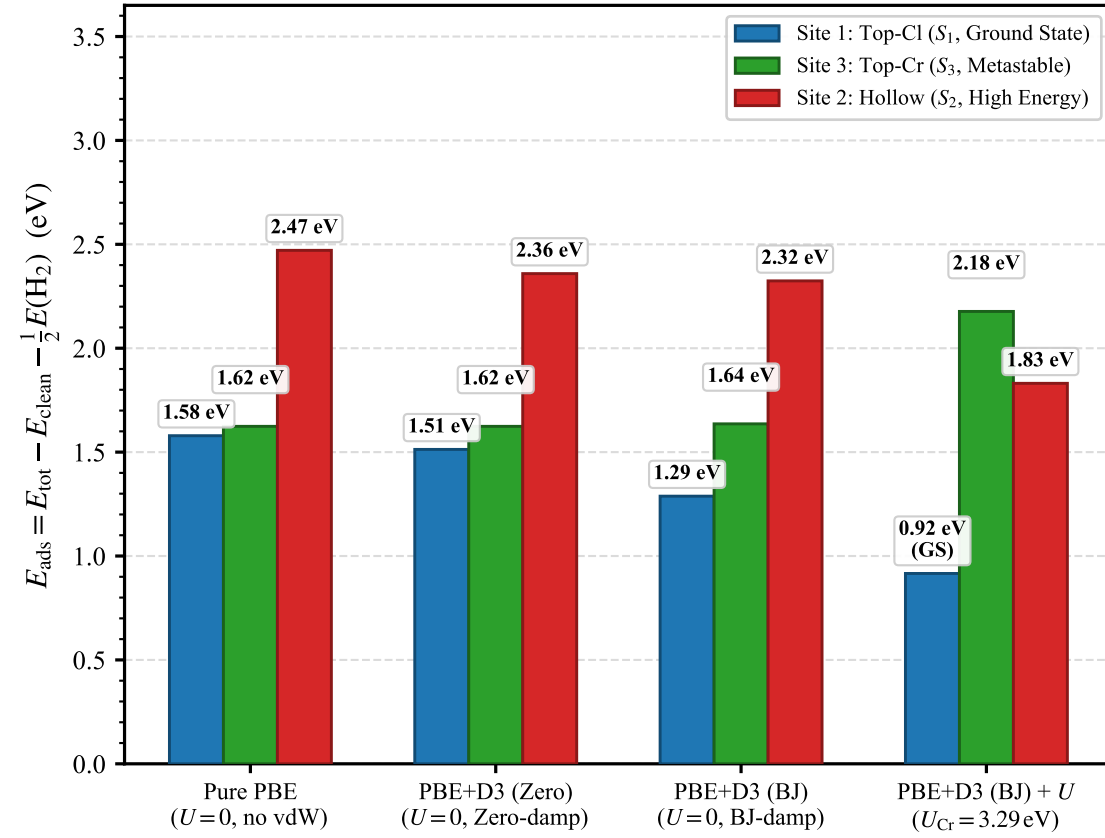
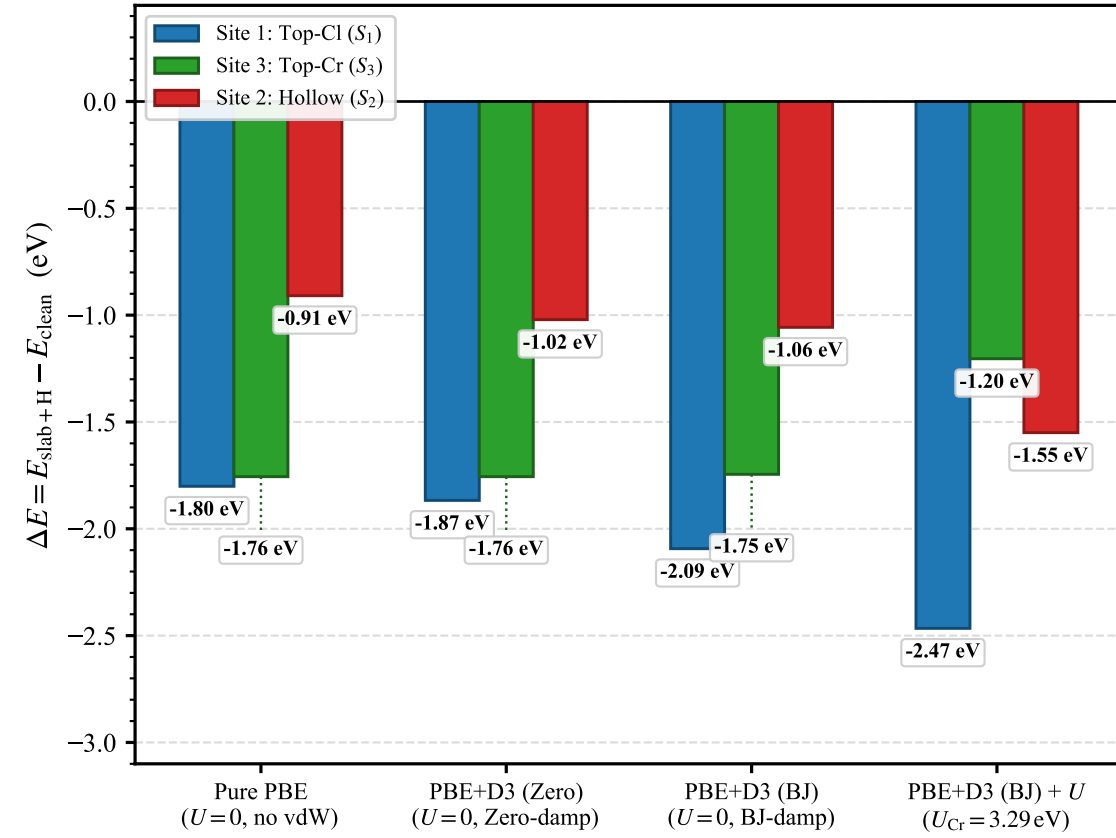


Comprehensive Hydrogen Adsorption & Catalytic Activation on 2×2 Monolayer CrCl_3 ($\theta=0.25$, $d_{\text{H-H}}=12.09 \text{ \AA}$)

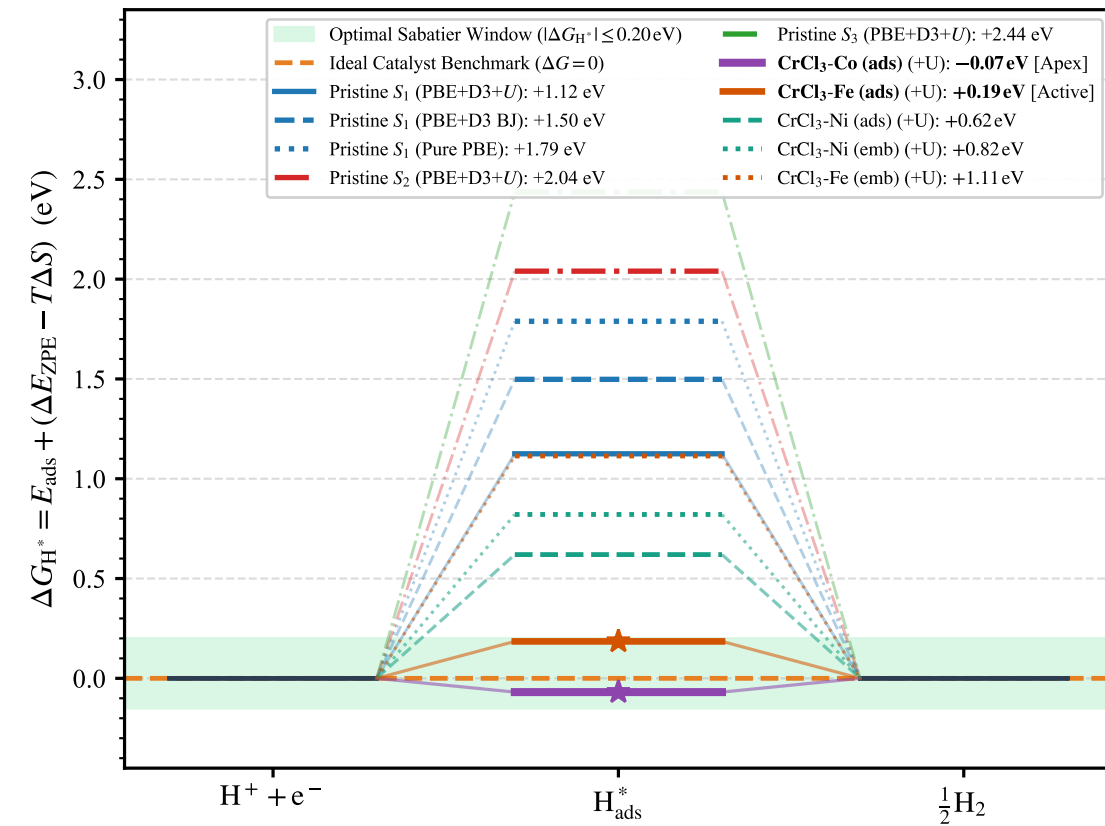
(a) 2×2 Hydrogen Adsorption Energy (E_{ads})



(b) Thermodynamic Binding Energy (ΔE)



(c) 2×2 HER Free Energy Profile: Pristine vs TM Activation



(d) Spin Polarization & Magnetic Compensation (2×2 Supercell)

